Akash Pandya

20 Harvard Street, Chatham, NJ 07928
973-255-0482 akashpandya1617@gmail.com akash-pandya-6b15152a6 akash-pandya.com

Education

University of Maryland - College Park **August 2023 – May 2027**
Bachelor of Science in Computer Science, Minor in Economics *GPA: 3.5*

Technical Skills

Languages: Java, Python, C++, C, SQL, JavaScript, Bash, R, HTML, CSS
Frameworks & Libraries: Spring Boot, Angular, TensorFlow, Scikit-learn, Pandas, PostgreSQL, NumPy, Matplotlib
Developer Tools: Git, GitHub, Jenkins, Maven, DBeaver, SonarQube, Jira, IntelliJ IDEA, VS Code, Kiro, Amazon Bedrock

Experience

Depository Trust and Clearing Corporation (DTCC) **June – August 2024, 2025, 2026**
Information Technology Intern *Jersey City, New Jersey*

Summer 2026

- Remediated 11 SQL injection vulnerabilities across 10 production components and 3 core data services using PostgreSQL and DBeaver, executing 40+ live database tests and resolving 3 additional critical functional regressions.
- Built a production Trade Information Warehouse (TIW) retrieval pipeline, indexing hundreds of settlement documents for vector-based semantic search with typical chatbot responses under 1 minute.

Summer 2025

- Supported migration of legacy Funds Only Settlement web components into DTCC's Real-Time Trade Matching application, improving application reliability and maintainability.
- Shipped 8+ Angular UI components and Java/Spring Boot REST APIs while maintaining 90% unit test coverage and resolving Jenkins CI/CD build failures.

Summer 2024

- Automated KPI test-result aggregation using SPL, eliminating 2+ hours of manual reporting per week and enabling more frequent performance monitoring.
- Streamlined Global Trade Repository alert analysis using SPL, reducing alert response time by 30% and generating automated daily reports for the Alerts Team.

Projects

Agentic AI Trading Platform | *OpenClaw, Multi-Agent Systems, Python, Flask* ***May 2026 – Present***

- Developed a 17-agent investment research system spanning fundamental, technical, macro, risk, and portfolio workflows to evaluate cross-market opportunities and generate trade recommendations.
- Automated 9 scheduled research and monitoring processes, combining financial data ingestion, trailing-stop logic, and 2-sigma anomaly detection for portfolio risk monitoring.
- Built a responsive Python/Flask portfolio dashboard tracking P&L, Sharpe and Sortino ratios, CAGR, drawdown, exposure, VaR/CVaR, win rate, and active positions for autonomous execution with optional human intervention.

Spotify User Data Pipeline | *Python, NumPy, Matplotlib, Scikit-learn, TensorFlow* ***March – May 2026***

- Built an end-to-end data science pipeline across Spotify's top 100 artists, cleaning malformed and inconsistent listener data before feature engineering, statistical analysis, and model training.
- Engineered trend-rate and listener-gap features and trained a sequential neural network that achieved 93.7% test accuracy in identifying artists with high growth potential.
- Ranked top 5 artists by predicted growth probability to identify investment candidates, using model outputs and statistical validation to support data-driven investment decisions.

AI in Higher Education | *Research Synthesis, Python, Google Forms, Google Sheets, Mendeley* ***January – May 2026***

- Contributed to a systematic research synthesis on generative AI in higher education by screening and coding peer-reviewed studies from ERIC, Scopus, and ACM Digital Library.
- Helped narrow 398 initial sources to 47 final studies through title, abstract, and full-text review using predefined inclusion and exclusion criteria.
- Supported analysis of student learning outcomes across critical thinking, writing, metacognition, and academic performance to identify patterns in AI-assisted learning.